MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG, NEW DELHI 110002

व्यापक परिचालन मसौदा

हमारा संदर्भ — सीईडी 09/टी – 21 **20 अगस्त 2026**

तकनीकी समिति — टिम्बर एवं टिम्बर स्टोर्स विषय समिति , सीईडी 09

प्राप्तकर्ता :

1. **सिविल इंजीनियरिंग विभाग परिषद, सीईडीसी के सभी सदस्य**
2. **टिम्बर एवं टिम्बर स्टोर्स विषय समिति, सीईडी 09 के सभी सदस्य**
3. **सीईडी 09 के अंतर्गत पैनल एंव वर्किंग ग्रुप के सभी सदस्य**
4. **रूचि रखने वाले अन्य निकाय।**

प्रिय महोदय/महोदया,

कृपया निम्नलिखित मसौदा संलग्न देखें:

प्रलेख संख्या	शीर्षक
सीईडी 09 (35180) WC	**सामान्य पैकेजिंग कार्यों के लिए वुड वूल — विशिष्टि का भारतीय मानक मसौदा** ***(दूसरा पुनरीक्षण)*** **(ICS: 79.040)**

कृपया इस मसौदे का अवलोकन करें और अपनी सम्मतियाँ यह बताते हुए भेजे कि यह मसौदा प्रकाशित हो तो इस पर अमल करने में, आपको व्यवसाय अथवा कारोबार में क्या कठिनाइयाँ आ सकती हैं।

सम्मतियाँ भेजने की अंतिम तिथि — 04 अक्टूबर 2026

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धन्यवाद।

भवदीय
ह-/
द्वैपायन भद्र
वै. 'एफ'/ वरिष्ठ निदेशक और प्रमुख (सिविल इंजीनियरिंग विभाग)

संलग्न: उपरिलिखित

भारतीय मानक ब्यूरो
BUREAU OF INDIAN STANDARDS

MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG, NEW DELHI 110002

WIDE CIRCULATION DRAFT

Reference — CED 09/T–21 **Dated — 20 August 2026**

TECHNICAL COMMITTEE — TIMBER AND TIMBER STORES SECTIONAL COMMITTEE, CED 09

ADDRESSED TO:

1. **All Members of Civil Engineering Division Council, CEDC**
2. **All Members of Timber and Timber Stores Sectional Committee, CED 09**
3. **All Members of Panels and Working Groups under CED 09**
4. **All other interested**

Dear Madam/Sir,

Doc. No.	Title
CED 09 (35180) WC	Draft Indian Standard Wood Wool for General Packaging Purposes — Specification (*Second Revision*) (ICS: 79.040)

Kindly examine the draft and forward your views stating any difficulties which you are likely to experience in your business or profession, if this is finally adopted as National Standard.

Last Date for comments — 04 October 2026

Comments if any, may please be made in the attached format and mailed to the undersigned at the above address or preferably through e-mail to ced9@bis.gov.in; nishikant.singh@bis.gov.in . The comments may preferably be shared in the prescribed template through the Manak Online portal at www.manakonline.in. Alternatively, the comments may be sent through the attached format for consideration by the BIS' Sectional Committee for necessary action.

In case no comments are received or comments received are of editorial nature, you will kindly permit us to presume your approval for the above document as finalized. However, in case comments, technical in nature are received, then it may be finalized either in consultation with the Chairman, Sectional Committee or referred to the Sectional Committee for further necessary action if so desired by the Chairman, Sectional Committee.

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Thanking You

YOURS FAITHFULLY
Sd/-
Dwaipayan Bhadra
Sc. 'F' / Senior Director and Head (Civil Engineering Department)

ENCLOSED— As Above

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(Please use A4 size sheet of paper only and type within fields indicated. Comments on each clause/subclause/table/fig etc. be started on a fresh box. Information in column 5 should include reasons for the comments, and those in column 4 should include suggestions for modified wording of the clauses when the existing text is found not acceptable. Adherence to this format facilitates Secretariat's work) {Please e-mail your comments to ced9@bis.gov.in, nishikant.singh@bis.gov.in

Doc. No.	Doc: CED 09 (35180) WC
Title	Draft Indian Standard Wood Wool for General Packaging Purposes — Specification (*Second Revision*) (ICS: 79.040)
Last Date for Comments	04 October 2026
Name of the Commentator /Organization	

Sl No.	**Clause/Sub-clause/Para No.**	**Comments/Suggestions**	**Modified Wording of the Clause**	**Reasons/ Justifications for the Proposed Changes**
(1)	**(2)**	**(3)**	**(4)**	**(5)**

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DRAFT INDIAN STANDARD

WOOD WOOL FOR GENERAL PACKAGING PURPOSES — SPECIFICATION

(*Second Revision*)

Timber and Timber Stores Sectional Committee, CED 09	Last Date for Comments: **04 October 2026**

FOREWORD

(*Formal clause to be added later*)

Wood wool consists of thin pliable narrow strips of wood. It provides a loose cushioning suitable for nesting individual stores of interposing between and around a number of stores, where its moisture retaining and other contaminating properties are tolerable, for example, in the packing of ceramics and glassware. It may also be employed as a 'filling' or 'floating' medium where the danger from moisture and other contamination cannot be harmful, for example, between an inner container (having optical, scientific, range-finding instruments, etc.) and an outer container.

The amount of cushioning provided by wood wool is an important factor to be determined. The methods of tests to estimate this property are included in this version.

This Indian Standard was first published in 1960 and subsequently revised in 1979. This standard has been taken up for revision with a view to bring its provisions in line with current practices. In this revision, the following modifications have been incorporated:

a) BIS certification marking clause has been updated;
b) References to various Indian Standards have been updated; and
c) The standard has been aligned with the latest style and format of Indian Standards, with editorial and structural changes to improve clarity and ease of use.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded-off value should be the same as that of the specified value in this standard.

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DRAFT INDIAN STANDARD

WOOD WOOL FOR GENERAL PACKAGING PURPOSES — SPECIFICATION

(*Second Revision*)

Timber and Timber Stores Sectional Committee, CED 09	Last Date for Comments: **04 October 2026**

1 SCOPE

This standard covers the essential requirements of wood wool for general packaging purposes, for example, for the packing of machinery and other engineering equipment, furniture, glassware and ceramics.

2 RFERENCES

The standard given below contains provisions, which through reference in this text, constitute provisions of this standard. At the time of publication the edition indicated is valid. All standards are subject to revision, and parties to agreement based on this standard are encouraged to investigate the possibility of applying the most recent editions of this standard:

IS No.	*Title*
IS 4905 : 2015 / ISO 24153 : 2009	Random Sampling and Randomization Procedures (*first revision*)

3 TIMBERS SUITABLE FOR WOOD WOOL

3.1 Wood wool shall be manufactured from coniferous timbers only. The following species of timber have been found to be suitable:

Standard Trade Name	*Botanical Name*	*Abbreviated Symbols*
Chir	*Pinus roxburghii* (Syn. P. LONGIFOLIA)	CHR
Cypress	*Cupressus torulosa*	CYP
Deodar	*Cedrus deodara*	DEO
Fir	*Abies* spp.	FIR
Kail	*Pinus wallichiana* (Syn. P. EXCELSA)	BPI
Spruce	*Picea smithiana* (Syn. P. MORINDA)	SPR

3.1.1 Timber used for the manufacture of wood wool shall be thoroughly sound and adequately dry. It shall be free, as far as practicable, from knots and cross grain.

4 REQUIREMENTS

4.1 General — Wood wool shall be in the form of a knit mass of tangled strands and shall be clean and free from foreign matter and dirt.

4.2 Dimensions of Strands

4.2.1 The dimensions of individual strands of wood wool shall be as follows:

a)	Width	Not less than 1.5 mm and not more than 3 mm
b)	Thickness	Not more than 0.33 mm
c)	Length	Not less than 250 mm in at least 75 percent of the supply. Strands less than 125 mm in length shall not exceed 10 percent including dust and small pieces under 25 mm (*see* **4.2.2**)

4.2.2 *Dust and Small Pieces* — When the wood wool is examined at the manufacturer's premises by the method described in Annex A, the amount of dust and small pieces under 25 mm in length in each bale shall not exceed 5 percent of the total weight of the bale.

5 SAMPLING

5.1 Scale of Sampling

5.1.1 *Lot* — In any consignment all the bales of wood wool produced from the same species of timber and under identical conditions of manufacture shall be grouped together to constitute a lot.

5.1.2 The number of bales to be selected from a lot shall depend upon the size of the lot and shall be in accordance with Co1 1 and 2 of Table 1.

Table 1 Scale of Sampling

(*Clause* 5.1.2)

Sl No.	Lot Size N	Number of Bales to be Selected n
(1)	(2)	(3)
i)	Up to 50	2
ii)	51 to150	3
iii)	151 and above	5

5.1.3 These bales shall be selected at random from the lot. In order to ensure randomness, all the bales in the lot may be arranged in a serial order and starting from any random bale, every rth bale may be included in the sample, Y being the integral part of $\frac{N}{n}$,

where

N = lot size, and

n = number of bales to be selected.

5.1.3.1 For other methods of ensuring randomness in selection of bales from a lot, IS 4905 may be referred to.

5.1.4 From each bale so selected, one sample of wood wool of about 720 g shall be obtained by mixing together equal quantities from the centre and each end.

5.2 Number of Tests

5.2.1 All the samples obtained as in **5.1.4** shall be tested for the requirements as given under **4** and **6**.

5.3 Criterion for Conformity — A lot shall be considered as conforming to the requirements of this standard if all the samples selected from it satisfy the various test requirements.

6 TESTS

6.1 Moisture Content — The moisture content of wood wool, when determined by the method described in Annex B, shall not be less than 5 percent and not more than 15 percent at the time of inspection.

NOTE — As the moisture content of wood wool is liable to change with changes in atmospheric conditions, it is emphasized that the term ‘at the time of inspection’ applies to the time when the wood wool is accepted by the inspecting officer or his representative.

6.2 *p*H Value — Where required by the purchaser at the time of placing the order, the PH value of the water extract determined by the method described in Annex C shall not be less than 4.5 and not more than 8.0.

6.3 Resistance to Fracture — Individual strands of wood wool, 150 mm long, conditioned to a constant mass at 65 ± 5 percent RH and 27 ± 2°C temperature shall not fracture when wound around a 12 mm diameter rod or when given at least three complete twists by holding one end in one hand and twisting the other end.

6.4 Tests for Estimating Cushioning Characteristics

6.4.1 *Compressibility Test* — Wood wool sample as obtained in **5.1.4** shall be tested as described in Annex D. The load required to compress the wood wool up to l/2 of the volume shall not be more than 70 kg and the residual compression after removal of load shall not be more than 60 percent of the total compression.

6.4.2 *Shock Absorption Test* — Wood wool sample as obtained in **5.1.4** shall be tested as described in Annex E. The rebound shall not be more than 10 percent of the height of drop.

7 PACKING

7.1 Wood wool shall be packed in bales of 25 kg or 50 kg each; the stated weight shall be the net weight exclusive of battens and similar accessories used for packing purposes.

8 MARKING

8.1 Each bale of wood wool shall be legibly and indelibly marked (stamped or stencilled) with the following particulars along with such other information as the purchaser may stipulate at the time of placing an order:

a) Manufacturer’s name, initials or recognized trade-mark;
b) Year of manufacture;
c) Net weight of wood wool in kg; and
d) The words ‘WOOD WOOL’.

8.2 BIS Certification Marking

The product(s) may be marked with Standard Mark as per the conformity assessment schemes governed by the provisions of the *Bureau of Indian Standards Act*, 2016 and the Rules and Regulations made there under. The details of conditions for the license may be obtained from the Bureau of Indian Standards.

ANNEX A

(*Clause* 4.2.2)

DETERMINATION OF DUST AND SMALL PIECES

A-l From the resultant mass as chosen under **5.1.4**, take about 400 g of wood wool, shake it in a bag and spread out on a clean surface. Examine visually and gravimetrically for dust and small pieces.

ANNEX B

(*Clause* 6.1)

DETERMINATION OF MOISTURE CONTENT

B-l A small sample of about 10 g, taken from the mass as chosen under **5.1.4**, shall be placed immediately in a tared metal or glass weighing container having a tightly fitting cover. A separate weighing container shall be used for each sample taken. The sample and its container shall then be weighed to the nearest 0.01 g. The cover shall remain tightly fitted until the weighing is completed. The cover is then removed and the container with sample inside shall be placed in an oven at a temperature of 100 to 105 °C for 24 hours or until the mass is constant (W_0). Each time the container is removed from the oven for weighing, the cover shall be put securely in place.

B-2 The moisture content shall be calculated as follows:

$$\text{Percentage of moisture} = \frac{W_1 - W_0}{W_0} \times 100$$

where

W_1 = mass of the sample before drying, and

W_0 = oven-dry mass of the sample.

B-3 Care shall be taken to prevent any change in moisture content between the sampling and the first weighing and also between removal from the oven and the subsequent weighing.

ANNEX C

(*Clause* 6.2)

DETERMINATION OF pH VALUE

C-l Take a small sample of log, from the resulting mass as chosen under **5.1.4**, add 400 ml boiling distilled water and stir thoroughly. Allow to stand for about 18 hours, taking suitable precautions against contamination by atmospheric impurities. Stir thoroughly and allow to stand further for an hour. Decant off 200 ml of the solution (filtering, if necessary, through a dry sintered glass filter and rejecting the first 25 ml of the filtrate) and determine the *p*H value by any suitable standard method.

C-2 Simultaneously, run a blank test for checking the accuracy of the method adopted.

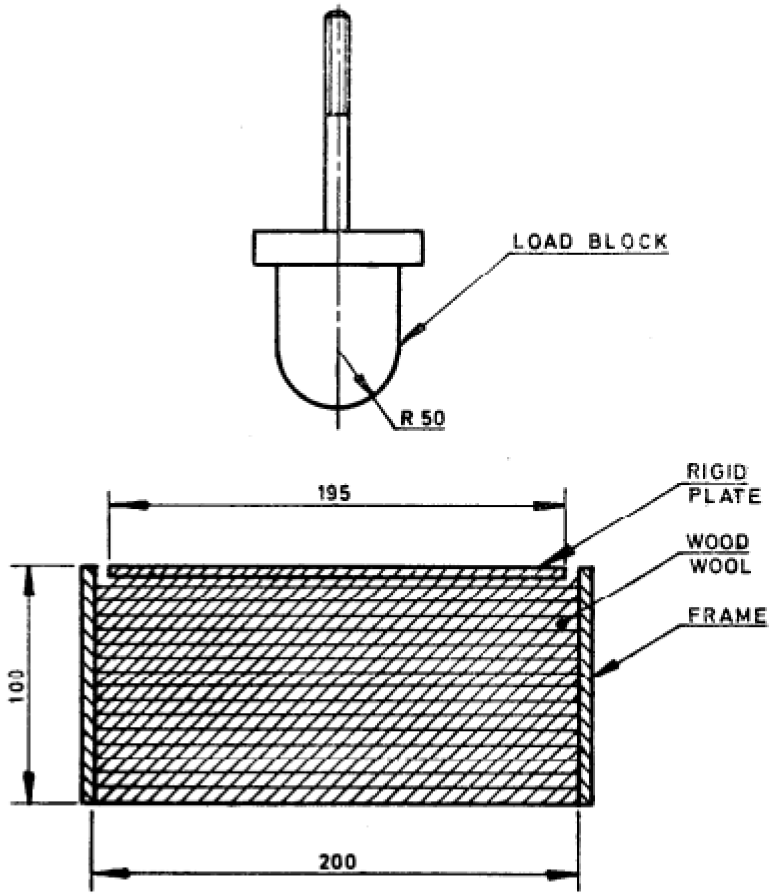
ANNEX D

(*Clause* 6.4.1)

COMPRESSIBILITY TEST

D-l PROCEDURE

D-l.1 From the sample of wood wool obtained as in **5.1.4**, 150 g wood wool strands shall be taken and conditioned at 65 ± 5 percent RH and 27 ± 2 °C temperature. The conditioned sample shall be filled in a frame of 200 × 200 × 100 mm in size in a manner as to simulate the general compactness in packaging. The frame shall be kept on the platform of a universal testing machine. A rigid plate of 195 × 195 mm in cross section shall be placed over the wood wool. The load shall be applied by means of hemispherical loading block of radius 50 mm (*see* Fig. 1) at the centre of the plate with a continuous and uniform rate of motion of moving head equal to 2 mm per minute and compression shall be noted correct to 0.1 mm by means of a scale fixed on the loading block. The test shall be continued till 50 mm compression (that is, wood wool is compressed to half of its volume) is reached. For this compression the required load shall be recorded. The load shall then be removed and the residual compression shall also be recorded.



All dimensions in millimetres.

Fig 1 Compressibility Test of Wood Wool

D-2 The load required to compress the wood wool to half of its volume and residual compression immediately after the removal of the load shall comply with the requirements of **5.4.1**.

ANNEX E

(*Clause* 6.4.2)

SHOCK ABSORPTION TEST

E-1 PROCEDURE

E-l.1 From the sample of wood wool obtained in **5.1.4**, 150 g wood wool strands shall be taken and conditioned at 65 f 5 percent relative humidity and 27 f 2°C temperature. The conditioned sample shall be completely and uniformly filled in a frame of 200 × 200 × 100 mm in size and a rigid plate of 195 × 195 mm in cross section shall be kept over the wood wool. A cubical block of 100 × 100 × 100 mm shall be fixed on one end of a rod. The rod shall be square in cross section and graduated in centimetres as shown in Fig. 2. The total weight of the block with rod shall be 1 kg. The cubical block with rod shall be dropped 10 times on the centre of the plate from a height of 75 cm with the help of suitable guides. The rebound of the rod shall be read at any convenient horizontal position by means of the guide at each drop.

E-l.2 The rebound shall not be more than 75 mm (that is, 10 percent of the height of drop) in all the 10 drops.

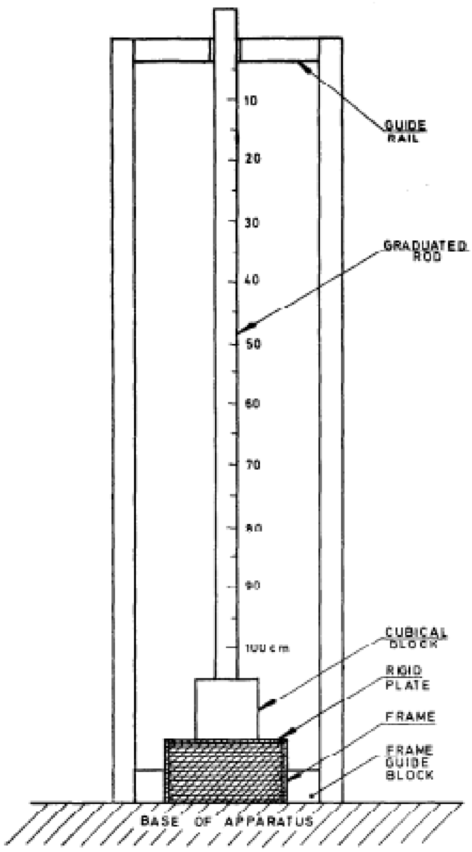


Fig. 2 Shock Absorbing Test of Wood Wool